

Akilan Sankaran

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EDUCATION

Harvard University | *Freshman* 2025 – 2029 (Expected)
Math 55; Physics 16; LifeSci 1A; ML-based Materials Research @ Kozinsky Lab (**XC Functionals, Neural Network Potentials**); ML/Econometrics Project Lead for Charles River Economics Labs and Machine Intelligence Community

Albuquerque Academy | *High School Student* 2018 – 2025
Weighted GPA: 4.68/4.00; Highest in Class. *Relevant Coursework*: College-level Biochemistry/Organic Chemistry.

SCIENTIFIC RESEARCH AND DEVELOPMENT EXPERIENCE

Computational Materials Research Intern, Sandia National Laboratories (U.S. DOE) 2024 – 2025

- Evaluated carrier capture recombination rates within GaN semiconductors; utilized first-principles Density Functional Theory (DFT) methodologies to study transition probabilities at point defects in generalizable fashion.
- Developed first Python-based simulation library to evaluate recombination rates under a transition-state theory.

Degradation of “Forever Chemicals” via Electronic Structure Calculations 2023 – 2025

- Harnessed semiclassical approximations, quantum-scale methodologies (DFT/MD), and Fourier analysis to develop and implement electronic-level modeling of several Per- and Polyfluoroalkyl Substances (PFAS).
- Demonstrated first mechanistic understanding of Vitamin B₁₂-catalyzed defluorination of several PFAS.

Research Science Institute (RSI) – Pilot-Wave Hydrodynamics Research (MIT) ([Preprint](#)) 2024 – 2025

- Studied macroscopic tunneling analogues for millimetric walking droplets in MIT’s Applied Mathematics Lab.
- Formulated high-fidelity, experimentally validated droplet dynamics model within variable-topography systems.

Transgenic *Drosophila* Line Generation via CRISPR knock-in Methodology 2024 – 2025

- Wet-lab (collab. Stanford); enhancer-trapped stable stock generation via multi-generation crossing scheme.
- Fluorescence imaging of serotonergic activity to assess transcription-factor binding after balancer-based selection.

Unlocking the *abc*’s of Number Theory 2021 – 2023

- Developed and analyzed alternative quality metrics to that of the *abc*-conjecture to enumerate high-quality triples.

TECHNICAL DEVELOPMENT PROJECTS

Slideo: An Integrated Approach for General Speech-to-Command Slide Generation Fall 2025

- As Team Lead, designed and developed an end-to-end, fully automated, AI-based plugin for presentation generation and customization with Google Slides. Integrated with Google Chrome and OpenAI Whisper API for general use.

Alternatives to Standard Topic Modeling Approaches 2022 – 2023

- Investigated tailored alternatives to traditional topic-modeling algorithms within natural language processing.

INTERNATIONAL AND NATIONAL-LEVEL SCIENTIFIC RECOGNITION

- Regeneron ISEF**: Two-time Top Mathematics and Chemistry Grand Awards (2022, 2024).
- Coolidge Scholar**: Full-ride, presidential scholarship (5 out of over 4000 students).
- Regeneron Science Talent Search (STS)**: Top 40 Finalist and sole Seaborg Speaker Award.
- Broadcom MASTERS**: Winner of the Samueli Foundation Prize (Top Award) (1 out of ≈ 3000).
- Steven H. Strogatz Prize for Mathematics Communication**: Top Winner, Social Media (2023).
- Research Science Institute (RSI)**: Top 5 oral presentation out of ≈ 100 students, selected from > 3000 .
- USA Math Talent Search**: Top 7 scorers nationally; two-time gold medalist; **AIME**: 4-time qualifier.

TECHNICAL SKILLS

Languages/Tools: Python, L^AT_EX, MATLAB, Mathematica, PyTorch, JavaScript, C++, VS Code, Git, Bash, Claude/Cursor

ML/Math: ODE/PDE’s, Fourier/functional analysis, optimization, measure/Lebesgue/representation theory; NLP, Kalman filters, numerical solvers, Speech-to-Text Models, Topic Models, LSTM, Bayesian processes, stochastic modeling

Computational Chemistry/Biology: *ab-initio* modeling, quantum chemistry codes (Quantum Espresso/LAMMPS/VASP), ASE/VESTA/Symbio, ML Force-Fields and Functionals, mechanistic reaction analysis, electron–phonon coupling, perturbation theory.

Wet Lab: EMSA, cell culture, mass spec, *Drosophila* CNS morphology/dissection, CRISPR protocol, PCR, balancer selection, organic synthesis (Fischer esterification, elementary reactions), reporter assays, immunostaining, NMR/FTIR, XRD, reporter quantification